

Vamshi Krishna Reddy Rajanala

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PROFESSIONAL SUMMARY

VLSI and embedded systems engineer with expertise in RTL design, FPGA implementation, and hardware verification. Proven ability to architect complex digital pipelines, optimize timing closure, and validate signal integrity in real-time systems. Strong foundation in circuit design, physical design, and board-level bring-up across analog and mixed-signal domains.

EDUCATION

Texas A&M University – Kingsville
Master of Science, Electrical Engineering

Aug 2024 - May 2026
GPA: 4.0 / 4.0

Jawaharlal Nehru Technological University, Hyderabad
Bachelor of Technology, Electronics and Communication Engineering

Jul 2019 - Jun 2023
GPA: 3.25 / 4.0

SKILLS

Physical Design / VLSI: timing closure, synthesis, place and route, Silvaco TCAD, VLSI fundamentals

RTL and Logic Design: Verilog HDL, SystemVerilog, VHDL (familiar), RTL design, microarchitecture/pipeline design, testbench development

PCB and Circuit Design: Cadence OrCAD, Cadence Allegro, schematic capture, PCB layout, LTSpice, analog circuit design, power management circuitry

Simulation and Debug: SPICE simulation, RTL simulation, waveform analysis, functional verification, post-synthesis debug

Lab and Bring-Up: oscilloscopes, multimeters, logic analyzers, spectrum analyzers, signal generators, board-level bring-up and validation, signal integrity validation

Protocols and Interfaces: I2C, SPI, UART (register-level implementation), AXI bus architecture (familiar)

Programming Languages: Python, C/C++, Embedded C, MATLAB, CUDA, TCL scripting for EDA automation

Platforms and OS: Xilinx Vivado, Linux/UNIX, Windows

WORK EXPERIENCE

Project Intern, Digital Systems Verification and Validation

Jun 2023 - Sep 2023

Advanced Systems Laboratory (ASL), Defence Research and Development Organisation (DRDO)

- Characterized fiber optic, coax, and twisted pair cable performance for a missile telemetry ground station prototype using OTDR, TDR, and VNA measurements across the S-band operating frequency range (2200 to 2290 MHz) to select the lowest-loss cable type.
- Built and validated the ground station signal chain (horn antenna, low-noise amplifier, telemetry receiver, rugged computer) and coordinated dual ground stations to maintain continuous line-of-sight coverage as the target moved.
- Wrote MATLAB code to convert raw receiver voltage output into usable telemetry parameter values, turning previously unusable raw data into meaningful measurements.
- Operated oscilloscopes, multimeters, logic analyzers, and spectrum analyzers to measure signal behavior and validate fidelity across telemetry system parameters, documenting procedures and results for repeatable use.

PROJECTS

RISC-V Processor RTL Implementation

- Implemented RTL for a RISC-V RV32I 5-stage pipeline (fetch, decode, execute, memory, writeback) with hazard detection and forwarding logic to resolve data and control hazards.
- Built directed Verilog testbenches achieving functional coverage closure across ALU, branch, and load/store operations; analyzed pipeline stall and flush behavior to optimize instruction throughput.

Pipelined Multiply-Accumulate Unit for Xilinx FPGAs

- Designed a 5-stage pipelined 32x32-to-64-bit multiply-accumulate unit in Verilog for Xilinx FPGAs, splitting a single-cycle 32x32 multiplier into four 16x16 partial products across pipeline stages to shorten the critical path.
- Retained the single-cycle baseline in the project for a like-for-like structural comparison against the pipelined version.
- Authored the Vivado XDC constraint set (generic clock and I/O delay budget, no board pins) and documented the timing-closure procedure to run against a target part: synthesis, timing summary, worst-slack path, and iterative tightening of the clock period.
- Verified functional correctness with a self-checking Verilog testbench driving 40 back-to-back operations, including a mid-stream accumulator clear, against a behavioral reference model. All 40 matched.

Planar P-N Junction Diode Modeling

- Modeled and simulated a planar P-N junction diode using Atlas and DevEdit, defining mesh constraints and doping profiles for P-type ($5 \times 10^{18} \text{ cm}^{-3}$) and N-type ($1 \times 10^{16} \text{ cm}^{-3}$) silicon regions.
- Validated I-V characteristics under forward and reverse bias against theoretical models, reinforcing semiconductor device physics fundamentals that underpin VLSI and physical design.

3-Stage Class AB Power Amplifier

- Designed and simulated a high-gain (60 V/V / 35 dB) power amplifier schematic on a dual-rail $\pm 10\text{V}$ supply using TIP41C/TIP42C transistors, then built and validated a breadboard prototype to confirm real-world circuit behavior.
- Diagnosed crossover distortion to its root cause and resolved it via diode-based biasing, verifying clean sinusoidal output through iterative SPICE simulation and physical prototype testing.

Solar-Based Electronic Defense Uniform

- Engineered a multifunctional embedded hardware system integrating a 5W flexible solar panel, ATmega16A microcontroller, real-time thermal management, and GPS/GSM location tracking.
- Designed power management and energy-harvesting circuitry; implemented low-level I2C/UART driver routines for peripheral integration.

Fire Fighting Robot using Raspberry Pi

- Built an autonomous Fire Fighting Robot on a Raspberry Pi 3B, programming Python GPIO logic to poll flame and gas sensors and automatically trigger a relay-driven water pump and buzzer alarm on detection.
- Developed ESP32-CAM firmware in C/C++ for live camera streaming and WebSocket-based remote control of dual DC motors with PWM speed control, enabling browser-based navigation and teleoperation.

Footstep Energy Harvesting System

- Developed a piezoelectric sensor array converting mechanical energy into regulated electrical energy for mobile device charging.
- Designed signal conditioning, rectification, and voltage-regulation circuitry for analog-digital boundary circuit design.

CERTIFICATIONS & ACHIEVEMENTS

- PCB Design: Cadence OrCAD and Cadence Allegro Expert Certification - Udemy
- FPGA Engineer: Verilog with Xilinx Vivado Design Suite - Udemy
- Verilog HDL: VLSI Hardware Design Comprehensive Masterclass - Udemy
- VLSI Design Certification - Simplilearn

- Embedded System Design - Simplilearn
- Digital Signal Processing Certification - Simplilearn
- Project Certification: Simulation and Measurement of Telemetry Receiving System - DRDO